

4.

(a)

For one observation (x_n, t_n)

$$t_n | x_n, w \sim \mathcal{N}(w^T \phi(x_n), \sigma_n^2)$$

$$\Rightarrow p(t_n | x_n, w) = \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp \left[-\frac{(t_n - w^T \phi(x_n))^2}{2\sigma_n^2} \right]$$

PRIOR:

$$p(w|\alpha) = \mathcal{N}(w|0, \alpha^{-1}I)$$

$$= \left(\frac{\alpha}{2\pi} \right)^{M/2} \exp \left[-\frac{\alpha}{2} w^T w \right]$$

$$\therefore p(t_n | x_n, w) = \mathcal{N}(t_n | w^T \phi(x_n), \sigma_n^2)$$

$$\text{and } p(w|\alpha) = \mathcal{N}(w|0, \alpha^{-1}I)$$

(b)

~~Let~~

$$\text{Let } D = \{(x_n, t_n)\}_{n=1}^N$$

Assumptions: ~~are~~ observations are iid

$$p(t | X, w) = \prod_{n=1}^N p(t_n | x_n, w)$$

$$= \prod_{n=1}^N \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp \left[-\frac{(t_n - w^T \phi(x_n))^2}{2\sigma_n^2} \right]$$

Taking ~~the~~ log,

$$= -\frac{1}{2} \sum_{n=1}^N \left[\log(2\pi\sigma_n^2) + \frac{(t_n - w^T \phi(x_n))^2}{\sigma_n^2} \right]$$

$$w_{MLE} = \arg \max_w \left[-\frac{1}{2} \sum_{n=1}^N \frac{(t_n - w^T \phi(x_n))^2}{\sigma_n^2} \right]$$

$$= \arg \min_w \left[\frac{1}{2} \sum_{n=1}^N \frac{(t_n - w^T \phi(x_n))^2}{\sigma_n^2} \right]$$

For MAP,

$$p(w|x, t) \propto p(t|x, w) p(w)$$

$$\Rightarrow \log p(w|x, t) = \log p(t|x, w) + \log p(w) + \text{constant}$$

WKT// $\log p(w) = \frac{\alpha}{2} w^T w + \text{constant}$

$$\begin{aligned} \therefore w_{\text{MAP}} &= \arg \max_w \left[-\frac{1}{2} \sum_{n=1}^N \frac{(t_n - w^T \phi(x_n))^2}{\sigma_n^2} - \frac{\alpha}{2} w^T w \right] \\ &= \arg \min_w \left[\frac{1}{2} \sum_{n=1}^N \frac{(t_n - w^T \phi(x_n))^2}{\sigma_n^2} + \frac{\alpha}{2} w^T w \right] // \end{aligned}$$

(c) Giv

$$E_D(w) = \frac{1}{2} \sum_{n=1}^N r_n (t_n - w^T \phi(x_n))^2$$

WKT// $= \frac{1}{2} \sum_{n=1}^N \frac{(t_n - w^T \phi(x_n))^2}{\sigma_n^2}$

$$\Rightarrow r_n = \frac{1}{\sigma_n^2}$$

Taking derivative, & equating to 0

$$0 = -\frac{1}{2} \sum_{n=1}^N r_n \phi_n (t_n - w^T \phi_n) [\phi_n = \phi(x_n)]$$

$$\Rightarrow \sum_{n=1}^N r_n \phi_n t_n = \sum_{n=1}^N r_n \phi_n \phi_n^T w$$

$$\Rightarrow w_{\text{ML}} = \left[\sum_{n=1}^N r_n \phi(x_n) \phi(x_n)^T \right]^{-1} \left[\sum_{n=1}^N r_n \phi(x_n) t_n \right]$$

Let $R = \begin{bmatrix} r_1 & 0 & \dots & 0 \\ 0 & r_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & r_N \end{bmatrix}$ $t = \begin{bmatrix} t_1 \\ \vdots \\ t_N \end{bmatrix}$ $\Phi = [\phi(x_1) \dots \phi(x_N)]$

$$\Rightarrow \text{Objective: } = \frac{1}{2} (t - \Phi^T w)^T R (t - \Phi^T w)$$

Differentiating & equating to zero,

$$\phi^T R (t - \phi^T w) = 0$$

$$\Rightarrow \phi^T R t = \phi^T R \phi^T w$$

$$\Rightarrow w_{ML} = (\phi^T R \phi^T)^{-1} \phi^T R t //$$